Design and Integration of a Modular Cyber-Physical System for Automated Waste Classification with a Collaborative Robot, Embedded Vision, and Cloud Traceability

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***Abstract*—The correct separation of recyclable waste at the point of disposal remains a largely manual task, with limited reliability and almost no traceability of who recycles what. This paper presents the design and integration of a modular cyber-physical system that automates the classification of three common packaging fractions: glass bottles, aluminum cans, and Tetra Pak containers. A collaborative robot equipped with a soft gripper manipulates the deposited items, an embedded vision service based on a lightweight convolutional neural network identifies the material, an industrial controller drives the reception mechanism, and an RFID interface identifies users, while a cloud layer records every classified item per user and publishes consolidated rankings that turn recycling into a gamified activity. The architecture is organized into three loosely coupled domains: the physical cabinet, the cloud services, and the user interfaces, coordinated by an edge orchestrator through the native protocol of each device. The design decisions behind this integration, the functional validation of the complete cycle on real waste items, and the current limitations of the prototype are discussed.**

***Index Terms*—cyber-physical systems, collaborative robotics, waste classification, embedded vision, cloud traceability, human-machine interface, gamification**

I. Introduction

Global municipal solid waste generation is expected to reach 3.4 billion tonnes per year by 2050, and a substantial fraction of recyclable material is lost because it is never separated correctly at the point of disposal [1]. Recycling chains depend on early and accurate separation: once glass, metal, and multilayer packaging are mixed and contaminated, recovery becomes more expensive and often economically unviable [1]. Manual sorting, whether performed by citizens at collection points or by workers at recovery facilities, is slow, inconsistent, repetitive, and offers no record of individual participation that could be used to encourage better habits.

Cyber-physical systems (CPS) integrate computation, networking, and physical processes under feedback loops in which embedded computers monitor and control physical actuators [2]. Although the CPS paradigm emerged mainly in manufacturing contexts [3], the same principles apply naturally to an urban recycling point: a machine that receives an object, verifies and classifies it, physically deposits it in the correct container, and registers the transaction for later analysis. Building such a machine is, above all, an integration problem, since it requires industrial control, robotic manipulation, machine perception, custom electronics, and Internet services to cooperate within one coherent architecture.

Such a recycling point is also a human–machine interaction problem: participation must be effortless for untrained users, and the system must return immediate, personally meaningful feedback if it is to influence recycling habits, which motivates combining contactless identification, a touchscreen interface, and game design elements [12]. Two recent developments make this integration practical today. Lightweight convolutional architectures [7] and embedded inference runtimes [9] allow image classification to run on low-cost single-board computers, and collaborative robots can operate near the public under bounded forces and speeds [10], [11], so the required perception and manipulation capabilities no longer demand industrial-scale infrastructure.

This paper describes the design, construction, and functional validation of a modular CPS that automates the classification of three packaging fractions: glass bottles, aluminum cans, and Tetra Pak containers. The system combines collaborative robotic manipulation, embedded machine-vision classification, industrial control of the reception mechanism, and RFID user identification with a cloud layer that records every classified item per user and publishes public rankings, turning recycling into a gamified activity. The prototype was assembled and demonstrated in a university robotics laboratory.

The contribution of this work is threefold. First, it proposes a three-domain modular CPS architecture (physical cabinet, cloud services, and user interfaces) for automated waste classification, with explicit, narrow interfaces between subsystems. Second, it documents the vertical integration of heterogeneous industrial and information technologies, from a collaborative robot and an industrial PLC to an embedded vision service, custom RFID hardware, and managed cloud services, through the native protocol of each device. Third, it introduces a per-user traceability and gamification scheme that links each physically classified item to a cloud record and a public ranking. The complete cycle is validated functionally on real waste items, and the integration lessons and current limitations

are discussed.

II. Related Work

Robotic waste sorting has been deployed industrially for more than a decade. An industrial recycling system combined multisensor perception with gantry manipulators to pick construction and demolition waste from a conveyor using machine learning [4]. On the perception side, public datasets such as TrashNet enabled the systematic study of image-based recyclable classification [5], and convolutional neural networks have since reported high accuracy on multiclass waste recognition tasks [6]. Most of these works concentrate on classification performance in isolation; the surrounding automation, user interaction, and data management are usually out of scope.

Lightweight convolutional architectures made it practical to run image classifiers on embedded boards [7], and end-user tools such as Teachable Machine lowered the barrier to training transfer-learning models without machine learning expertise [8], exporting them to TensorFlow Lite for edge deployment [9]. This work exploits exactly that route to obtain its embedded classifier.

Collaborative robots are designed to share space with people under bounded forces and speeds [10], with safety requirements consolidated in ISO/TS 15066 [11]. Their intrinsic safety and ease of programming make them attractive for service applications close to the public, such as the waste reception cell presented here.

Finally, gamification, understood as the use of game design elements in non-game contexts [12], has been widely applied to motivate sustainable behavior. Commercial reverse vending machines reward users for returning containers, but they tend to be proprietary platforms, restricted to a single material stream and closed to extension. The system presented in this paper differs in that it documents an open, modular integration that spans from the industrial control layer to per-user cloud traceability on a collaborative robot platform, a combination that is rarely reported as a whole.

III. System Architecture

A. Domain Decomposition

The architecture is organized into three loosely coupled domains, shown in Fig. 1: (i) the *robot cabinet*, which contains all the physical infrastructure (structure, robot, gripper, PLC, door mechanism, sensors, local computers, and networking); (ii) the *cloud services*, which host the REST API, the real-time database, and the public web application; and (iii) the *user interfaces*, which include the local touchscreen HMI with RFID identification and the public ranking website. Each domain exposes narrow, well-defined interfaces (HTTP endpoints, PLC memory bits, robot program names), so that modules can be developed, tested, and replaced independently. In Fig. 1, each arrow points from the module that initiates the interaction to the module that serves it, except for the display and RFID links, where the arrow indicates the direction of the data flow.

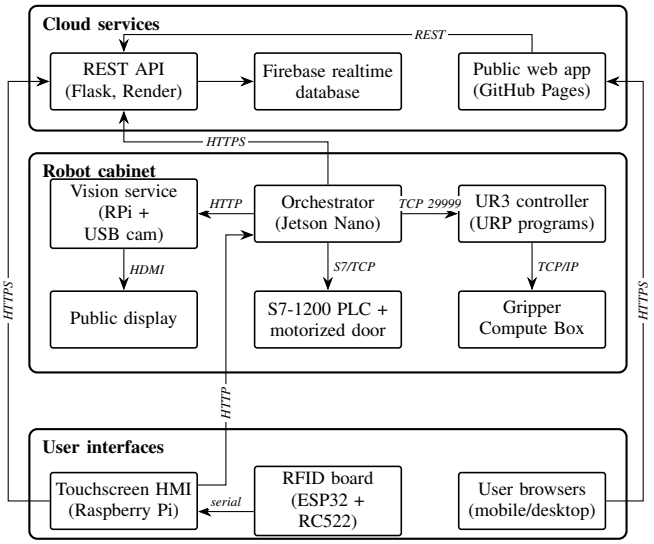


Fig. 1. Logical architecture: three loosely coupled domains and the native protocol used at each interface.

B. Network Topology and Protocols

Inside the cabinet, the devices form a star topology around an Ethernet switch with static IPv4 addressing in a private /24 segment; the switch uplinks to the campus network for Internet access. The edge orchestrator is an NVIDIA Jetson Nano developer kit (quad-core ARM Cortex-A57, 4 GB of RAM), while the vision service and the HMI run on separate Raspberry Pi single-board computers; all local services are implemented in Python. Instead of imposing a unified middleware, the integration deliberately uses the native protocol of each device: the S7 protocol (through the Snap7 open-source library [13]) for the PLC, the dashboard server on TCP port 29999 for loading and starting robot programs [14], and HTTP/REST for the vision service, the HMI, and the cloud. This choice kept every vendor stack unmodified and reduced the integration surface to a small set of documented calls. Fig. 2 summarizes the resulting topology together with the role of each node and the protocols it speaks: every cabinet device connects to the switch over wired Ethernet, while the camera, the public display, and the RFID board attach to their host computers through USB, HDMI, and serial links, respectively.

The alternative of imposing a unified middleware, such as ROS 2 [16] or an MQTT or OPC UA integration bus, was considered and discarded: none of the industrial endpoints speaks those protocols natively, so the PLC, the gripper Compute Box, and the robot dashboard would each have required a custom bridge to be written, versioned, and maintained on top of the vendor stack it wraps, adding a layer of indirection without removing any of the underlying protocols. The accepted trade-offs are that the orchestrator is a single point of failure and that coordination is sequential rather than event-driven; scaling to multiple cells or richer telemetry would justify revisiting a message-oriented middleware, but for a single cell with one robot and one PLC, native protocols minimized both the

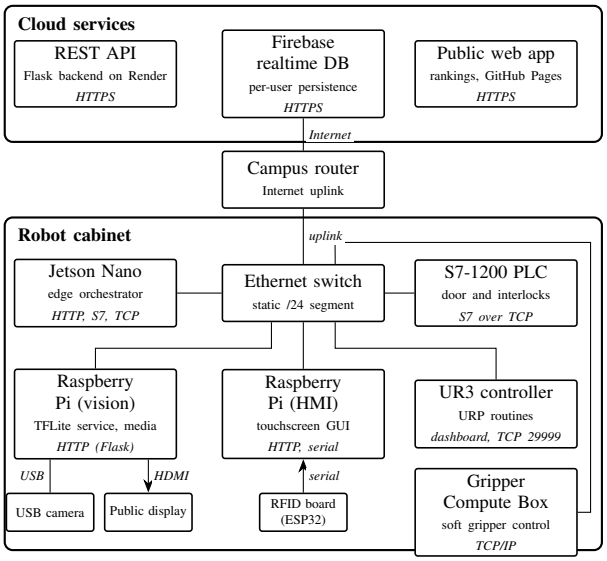


Fig. 2. Physical network topology: role and protocols of every node. Unlabeled links are wired Ethernet.

integration effort and the failure surface.

IV. Subsystem Design and Integration

A. Cabinet Structure and Robotic Cell

The cabinet is built from 40×40 mm aluminum extrusion profiles with a footprint of 1000×920 mm and a height of 2100 mm; Fig. 3 shows the integrated prototype in side, front, and top views. The UR3 is mounted at the top of the frame, centered along the width, so that its workspace covers both the reception tray and the classification containers. Three bins sit at the rear on a reinforced shelf; profiles at 45° and two caster-terminated legs stiffen the frame against the cantilevered loads. A 50-inch display above the bins shows public feedback during operation. The front of the cabinet is enclosed with 3-mm acrylic and PVC panels, separating users from the robot workspace; combined with the bounded forces of the collaborative arm [10], [11], this provides two independent layers of protection: the enclosure keeps users out of the workspace, and the force limits bound the severity of any residual contact through the reduced reception aperture.

For manipulation, the UR3 carries an OnRobot SG soft gripper, whose elastomer membrane adapts to variable geometries and fragile surfaces such as glass bottles. The gripper is managed by its Compute Box, connected to the cabinet switch over Ethernet and exposed to the robot controller as scriptable commands; grasping and releasing are performed by commanding target membrane widths from within the robot programs.

All trajectories are encapsulated as URP programs stored in the robot controller: an observation routine that places the camera viewpoint over the reception tray, one collection routine per material class, and one deposit routine per container. The orchestrator never commands joint motions directly during normal operation; it only selects which program to load and play. This decoupling keeps the kinematics where they are best maintained (the teach pendant) and the decision logic where it is best versioned (the orchestrator code).

TABLE I
Training Dataset Composition

Class	Images	Variability captured
Glass	49	Orientations of transparent bottles
Can	83	Colors, frontal/lateral/top views
Tetra Pak	79	Fill levels and wear
Background	10	Empty tray, irrelevant objects

B. Waste Reception Module

Objects enter the cabinet through a 170×330 mm aperture closed by a horizontally sliding door, which runs on V-slot wheels and is driven by a belt-pulley transmission coupled to a NEMA 17 stepper motor through a microstepping driver. A Siemens S7-1200 PLC (CPU 1215C) generates the pulse and direction signals for the driver from a ladder program, and the orchestrator commands the door by writing two memory bits (open and close) over the S7 protocol and reading them back to confirm the state. A guide piece on the reception tray acts as a poka-yoke: it shows the user where to place the object, constraining its position to the region expected by the robot collection routines.

Fig. 4 shows the wiring of the power and safety circuit of the module. Safety is implemented as a hardwired chain that is independent of all software: a normally closed emergency stop and a normally closed stop button feed a normally open start button that energizes an industrial relay with a self-holding auxiliary contact. The relay contacts switch the 127 VAC strip that powers the robot, the gripper Compute Box, and the local network and computing equipment, together with the 24 VDC supply of the stepper driver, so pressing the emergency stop removes power from every actuator regardless of the state of the PLC, the orchestrator, or the network. Control logic (24 VDC) and power loads are kept on separate circuits throughout the panel.

C. Embedded Vision and Classification

A USB camera observes the reception tray from the viewpoint defined by the robot observation pose, and is attached to a dedicated Raspberry Pi that runs the classification service. The classifier was obtained by transfer learning with Teachable Machine [8], which fine-tunes a lightweight convolutional backbone [7], and was exported to TensorFlow Lite [9] for embedded inference. Four classes are distinguished: glass, can, Tetra Pak, and a background class (“bg”) that captures empty scenes and non-target objects. Table I summarizes the training dataset, which was captured against a dark background under controlled illumination, varying object pose, color, and wear.

The service exposes a single HTTP endpoint that returns the detected label as JSON. To reduce the effect of transient misclassifications, the reported label is the statistical mode of several consecutive predictions taken within a short capture

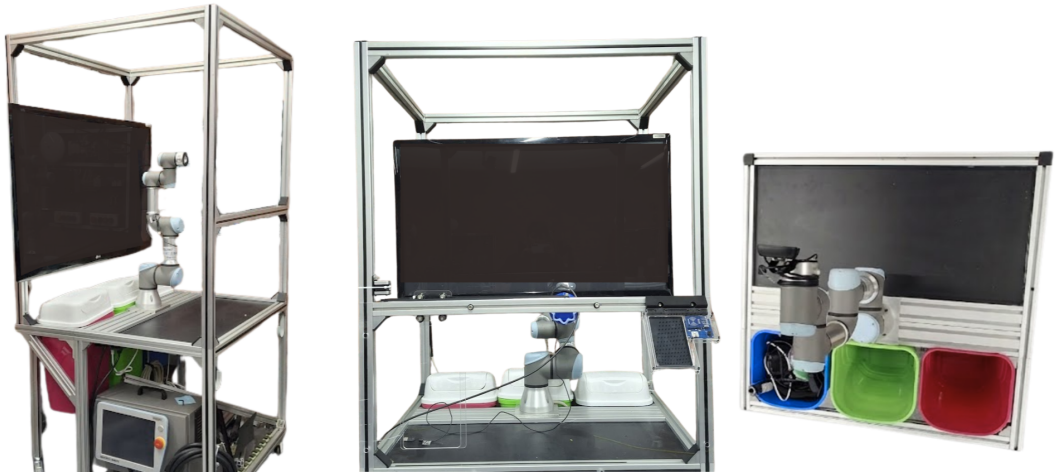


Fig. 3. Integrated prototype of the robot cabinet: side, front, and top views.

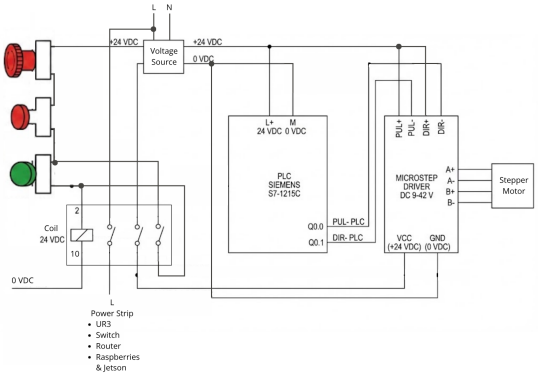


Fig. 4. Power and safety wiring of the waste reception module.

window, rather than a single-frame decision. When the result is "bg", the orchestrator aborts the cycle and returns the robot to its home pose, which prevents spurious manipulation of invalid objects; this rejection path is what allows the perception service to act as a gate for the actuation of the cell.

D. User Identification and Local HMI

User identification is handled by a custom single-layer PCB designed around an ESP32-WROOM-32U module and an RC522 13.56-MHz RFID reader connected over SPI, shown in Fig. 5. The board includes a buzzer for audible confirmation, two status LEDs, a user switch, and a 4-pin UART header for programming and debugging. Design rules followed embedded RF practice (a keep-out region under the ESP32 antenna, short parallel SPI traces, and hand-solderable passive footprints); the board was fabricated professionally and assembled and tested in the laboratory.

The local HMI runs on a Raspberry Pi with a 7-inch touchscreen mounted on the cabinet front at -45° for comfortable interaction; Fig. 6 shows the touchscreen and the RFID reading zone. The graphical application, developed in Tkinter, implements three flows: a *login* flow, where scanning a card retrieves the user name and accumulated score; an *account creation* flow, where a new card identifier is associated with a name and pushed to the backend; and an *anonymous recycling* flow, which allows participation without scoring. After a successful login, the user triggers the classification cycle from the screen, and the interface reports progress, errors, and the points obtained.

E. Edge Orchestration

The Jetson Nano runs a Flask server and an event-driven main loop that coordinate the complete cycle. The HMI signals the start of a deposit and binds the active user session through two HTTP requests to this server. When the start event arrives, the orchestrator first sends the UR3 to its home pose and commands the PLC to open the reception door; it then loads and plays the observation routine, queries the vision service, and closes the door once the windowed classification is available. If the label corresponds to a valid material, the corresponding collection and deposit routines are loaded and played through the dashboard interface, and when both a user session and a completed routine have been registered, the orchestrator posts the point increment to the cloud API and returns the robot to home. Every stage is guarded with timeouts and exception handling, so a failure in any link (robot telemetry loss, vision timeout, PLC disconnection) degrades to a safe abort that closes the door and returns the robot to home. While idle, the orchestrator pushes promotional media to the public display through the vision node, and during collection it streams the robot point of view to the same screen, intended to reinforce user engagement.

F. Cloud Services and Traceability

The backend is a Flask REST API deployed on Render, a platform-as-a-service provider, with credentials injected through environment variables and permissive CORS so that heterogeneous clients (robot, HMI, web) can reach it. Persistence uses the Firebase Realtime Database [15], a managed cloud service, with a simple per-user schema: name, creation

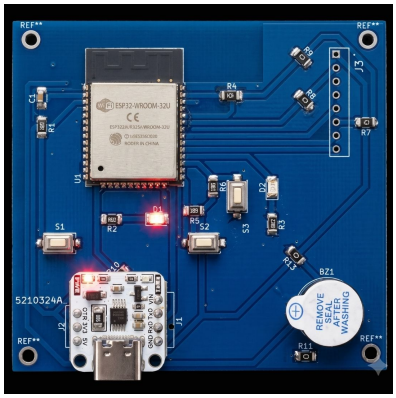


Fig. 5. Custom RFID identification board.



Fig. 6. Local HMI: touchscreen and RFID reading zone.

date, and one counter per material class. The API exposes endpoints to create users, query a profile, increment a counter for a validated deposit, and retrieve top-100 rankings by total and by material, with input validation and explicit error responses.

A public single-page web application, served through GitHub Pages, consumes these endpoints and presents each participant with their accumulated counts and position in the rankings. In the intended flow, every increment originates from a physically completed robot cycle, so the database constitutes a traceable record that links users, materials, and timestamps; the rankings convert that record into a gamification incentive [12].

V. Functional Validation and Discussion

A. Link-Level Integration Tests

Integration followed a bottom-up strategy: every interface was validated in isolation with short test scripts before composing the full cycle, writing and reading back the PLC door bits over S7; commanding UR3 joint motions with a 0.02-rad arrival tolerance and telemetry-loss guards; loading and starting URP programs through the dashboard server; and requesting classifications over HTTP. These tests located configuration faults (addressing, program paths, timeouts) early and produced a small regression suite that remained useful during the final assembly. The test scripts, code, and trained model will be released upon acceptance.

TABLE II
Detection Results (Ten Trials per Material)

Material	Detections	Mean conf.	Failure mode
Can	10/10	0.98	—
Glass	10/10	0.99	—
Tetra Pak	8/10	0.81	2× labeled glass (≈0.64)

B. End-to-End Operation

The integrated prototype executed complete cycles on real waste items of the three target fractions during laboratory demonstration sessions. In a representative cycle, a user authenticates at the HMI with an RFID card and requests a deposit from the touchscreen; the motorized door opens and the user places a glass bottle on the reception tray; the robot moves to the observation pose, the embedded classifier returns the glass label from its windowed decision, and the door closes; the UR3 then collects the bottle with the soft gripper and releases it into the glass bin, the orchestrator posts the increment to the cloud API, and the user's updated glass counter appears in the public ranking before the cabinet returns to idle.

The classification stage was then evaluated with ten deposit trials per material, recording in each trial the label reported by the windowed decision and the confidence score that the classifier assigned to it. The deposited items varied between trials as far as each stream allows: cans of different brands and colors, 600-ml glass bottles of two brands, and Tetra Pak containers of a single commercial type; the object position was similar in all trials by design, fixed by the poka-yoke tray guide of Section IV-B. Two quantities summarize the outcome, reported in Table II: the detection success rate, the fraction of trials whose reported label matched the deposited material, and the mean confidence, the average of the score associated with the reported label over the ten trials. Cans and glass bottles were detected in all ten trials, with mean confidences of 0.98 and 0.99, respectively. Tetra Pak containers were detected in eight of ten trials with a mean confidence of 0.81; in the two failed trials the classifier reported the glass label with a confidence of approximately 0.64, markedly lower than that of the correct detections. Overall, 28 of the 30 deposited items were classified correctly. The gap between the confidence of correct and incorrect labels suggests a practical refinement: a minimum-confidence threshold below which the cycle would abort in the same way as a background detection, converting residual misclassifications into safe rejections.

During these sessions, the background class rejected empty scenes and non-target objects, preventing spurious robot motion, and the windowed-mode decision visibly reduced label flickering. The hardwired safety chain was exercised as well: pressing the emergency stop removed power from the robot, gripper, and door driver as designed.

C. Discussion and Limitations

The clearest benefit of the architecture was the degree of independence between modules. Because each subsystem hides behind a narrow interface, the vision model can be retrained and redeployed without touching robot routines, deposit trajectories can be re-taught on the pendant without rebuilding the orchestrator, and the cloud schema evolved during development without affecting the cabinet. The subsystems were in fact developed in parallel by different team members and integrated late with limited rework, which we attribute to the protocol-level decoupling described in Section III.

Beyond modularity, the demonstration sessions support the central claim of this work: that the combination of an embedded perception service, a collaborative manipulator driven as a program selector, and a per-user cloud record is sufficient to automate attended waste classification at the point of disposal. Each element covered the role the architecture assigns to it. The embedded classifier gated the actuation of the cell, correctly detecting 28 of the 30 deposited items in the evaluation trials and rejecting invalid objects through the background class; the encapsulated URP routines completed collections and deposits on the three target fractions; and every completed cycle produced a cloud record that was immediately visible in the public ranking, closing the loop between a physical deposit and its digital trace. What remains to be established is how these figures hold under uncontrolled conditions and larger samples.

The current prototype also has clear limitations. The training dataset is small (221 images) and was captured under controlled illumination, so robustness to lighting changes and unseen packaging variants is limited; the Tetra Pak fraction, the weakest in the evaluation and tested with a single commercial type, is the natural priority when enlarging it. The validation reported here is functional and, beyond the thirty detection trials, qualitative: a statistically rigorous campaign measuring per-class accuracy under varied illumination, grasp and deposit success, and cycle times remains to be carried out. The cycle assumes a single object per deposit and uses fixed grasp poses, so items must rest in the region defined by the tray guide. Finally, the system depends on network availability, and the free-tier cloud deployment introduces cold-start latency on the first request. Moreover, with permissive CORS and no origin authentication on the increment endpoint, the cloud record is trustworthy only within the intended flow; securing the robot-to-cloud channel with per-device credentials remains pending work.

VI. Conclusions and Future Work

This paper presented the design and integration of a modular cyber-physical system that automates the classification of glass, cans, and Tetra Pak containers, combining collaborative manipulation, embedded vision, industrial door control, RFID user identification, and cloud traceability with gamified feedback. The three-domain architecture and the use of native device protocols kept the integration surface small and allowed subsystems developed in parallel to be composed into complete, automatically chained operation cycles on real waste items. The functional validation supports the claim stated in the Introduction: embedded perception gated the actuation of the cell, correctly detecting 28 of the 30 items deposited in the evaluation trials, encapsulated robot routines completed the physical sorting, and every completed cycle left a per-user cloud record.

Future work will proceed along four lines: enlarging and diversifying the training dataset and evaluating the classifier with formal metrics under uncontrolled illumination; instrumenting the orchestrator to log cycle times and outcomes for a quantitative performance campaign; adding object pose estimation to relax the fixed-grasp assumption; and extending the material set and the gamification mechanics supported by the cloud layer.

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